UNIVERSITY
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Bachelor's thesis

Foreign Exchange Time Series and Autoregressive Models

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Abstract

Financial time series are known to exhibit non-stationary behaviour and structural change. In particular, foreign exchange tick data may exhibit short-lived price runs and serial dependence. This report introduces and explores EUR/USD tick-by-tick data, describing the data format and the construction of pre-averaged prices and returns. I then introduce autoregressive models of order one, and fit such models to the data. The results of this fit illustrate the need for more flexible models in order to capture the short-term time-varying dependence in the data.

1 Introduction

Financial time series are well known to exhibit non-stationarity, volatility clustering, and structural change. Return autocorrelation at daily or lower frequencies is typically weak, but there is some empirical evidence of significant serial dependence at very short horizons. In foreign exchange (FX) markets, tick-by-tick prices are influenced by market microstructure effects. These effects can cause short-lived price runs and time-varying dependence structures.

The goal of this report is to explore tick-by-tick data from FX markets, specifically the EUR/USD exchange rate. The report explains how mid-prices and returns are constructed, and briefly introduces autoregressive models of order one (AR(1)). Such AR(1) models are then fitted to FX tick data split into 5-minute intervals, to assess the need for more flexible models.

The report is structured as follows. In Section 2, the FX tick data are presented. I explain the structure of the data, and introduce ways of constructing pre-averaged prices and returns. Section 3 provides a brief review of time series analysis, and presents AR(1) models, which are fitted to data in Section 4. Finally, I provide a brief plan for the project in Section 5, and give acknowledgements in Section 6.

2 Foreign Exchange Data

The foreign exchange (FX) tick-by-tick data used in this report is obtained from https://www.histdata.com/, which provides freely available tick and intraday FX data. In this report I consider the EUR/USD exchange rate. Each data entry consists of a Date-Time stamp, bid quote, ask quote, and volume. Table 1 contains the first 5 rows of the dataset, visualizing the structure of the data.

DateTime Stamp	Bid Price	Ask Price	Volume
2026-01-01 17:04:01.135	1.17387	1.17532	0
2026-01-01 17:04:37.436	1.17414	1.17531	0
2026-01-01 17:05:00.073	1.17414	1.17517	0
2026-01-01 17:05:00.283	1.17449	1.17523	0
2026-01-01 17:05:01.142	1.17449	1.17518	0

Table 1: Example of FX tick-by-tick data, downloaded from https://www.histdata.com/.

Let B_t and A_t denote the bid and ask prices at tick time t. Following common practice in high-frequency econometrics, I use the mid-price

$$P_t = \frac{B_t + A_t}{2}$$

as a proxy for the transaction price. Figure 1 shows an example of the EUR/USD mid-price over a single trading day, constructed in this way.

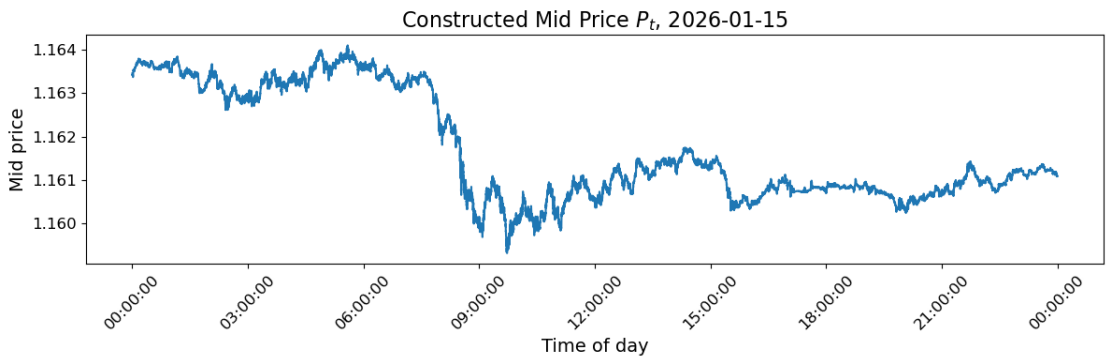

Figure 1: EUR/USD mid-price on January 15th 2026.

2.1 Pre-Averaging and Returns

Let P_t denote the observed mid price at tick time t. The observed price can be decomposed as

$$P_t = P_t^* + \eta_t,$$

where P_t^* is the latent efficient price and η_t represents the microstructure noise. To reduce the impact of noise, one can apply pre-averaging as discussed in Mykland and Zhang 2016. In this report, I consider a local average over k consecutive ticks. The pre-average price is defined as

$$\overline{P}_t = \frac{1}{k} \sum_{j=0}^{k-1} P_{t-j},$$

where t indexes the last tick in each averaging block. Using these pre-averaged prices, one can construct returns as

$$r_t = \overline{P}_t - \overline{P}_{t-1}.$$

These returns can then be modelled by an AR(1) model, as discussed in Section 3.2.

3 Background

3.1 Time Series

A time series is a collection of random variables over time, commonly denoted by $\{r_t\}$ (Tsay 2005).

A central concept in time series analysis is stationarity. The literature distinguishes between *strict* and *weak* stationarity. For most applications, weak stationarity is of interest, see Tsay 2005, p. 25.

Definition 3.1. A time series $\{r_t\}$ is weakly stationary if (a) $\mathrm{E}(r_t) = \mu$, which is a constant, and (b) for every integer l, $\mathrm{Cov}(r_t, r_{t-l}) = \gamma_l$, which only depends on l.

Weak stationarity implies that a time series has constant mean, and that the covariance between entries only depends on the lag l, not the time t. In other words, the covariance and mean are time-invariant. The assumption of weak stationarity is essential for the autoregressive model discussed in section 3.2.

3.2 Autoregressive Models

Autoregressive models provide a simple way of modelling time series data, by assuming serial dependence between entries.

Definition 3.2. An autoregressive model of order one (AR(1)) for a time series $\{r_t\}$ is given by

$$r_t = \mu + \phi r_{t-1} + \epsilon_t,$$

where μ is the unconditional mean return, ϕ measures the linear serial dependence and ϵ_t is a normally distributed innovation term with mean 0 and variance σ^2.

As discussed in Tsay 2005, p. 33, weak stationarity is both a sufficient and necessary condition for AR(1) models. AR(1) models assumes constant linear dependence ϕ, and therefore provide a useful reference point when comparing more flexible models in later work.

4 Results and Discussion

To check the assumption of constant serial dependence, I choose an arbitrary day and construct returns as described in Section 2.1, with $k = 20$ ticks in each pre-averaging block. The data is then split into 5-minute intervals, and I fit an AR(1) model on each interval. Figure 2 shows a plot of the estimated ϕ's as a function of time. As seen in the plot, the linear dependence is not constant. This motivates the need for more flexible models in order to capture the short-term time-varying dependence mentioned in Section 1.

5 Plan for the Project

The plan for the project is as follows. First, I will continue the literature review and exploratory data analysis of FX tick data. I will then try to model regime changes in FX tick data by Hidden Markov Models, AR(p) models and Online Kernel-Based methods. I will then compare and report the performance of these models. My project advisor and I have scheduled weekly meetings, where we will discuss the project progress.

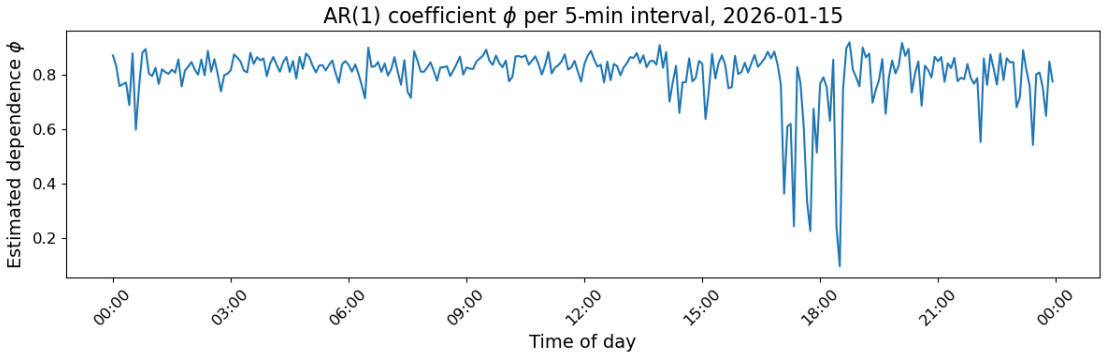

Figure 2: Linear dependence ϕ for each 5-minute interval on January 15th 2026. The linear dependence is not constant.

6 Acknowledgements

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References

Mykland, P. A. and Zhang, L. (2016). 'Between data cleaning and inference: pre-averaging and robust estimators of the efficient price'. English. In: *J. Econom.* 194.2, pp. 242–262. DOI: 10.1016/j.jeconom.2016.05.005.

Tsay, R. S. (2005). *Analysis of financial time series.* English. 2nd ed. Wiley Ser. Probab. Stat. Hoboken, NJ: John Wiley & Sons. DOI: 10.1002/0471746193.